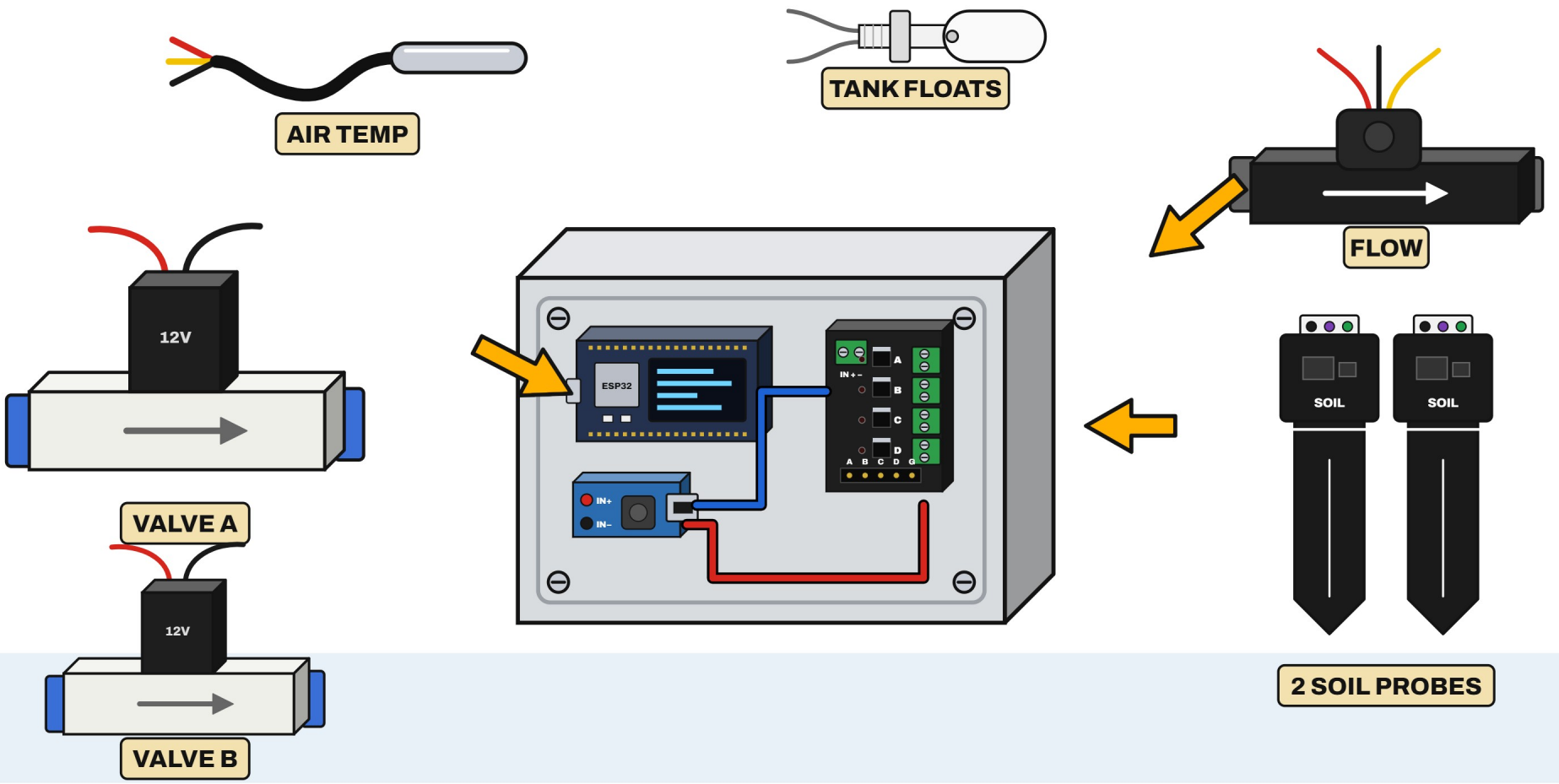
BUILD BOOK · ESP32 + 3 × 12V VALVES

Greenhouse Controller

25 BUILD STEPS

19 PIECES · ORDER #121677 + VALVE C



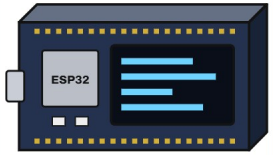
⚠ Adult check at drilling, soldering and first 12V power-up

⚠ Low voltage only. The 230V plug stays outside the box.

Parts in your PiShop box

Order #121677 · match every piece before you start

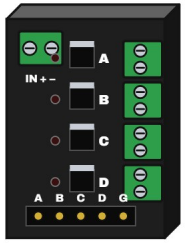
A **1x**



ESP32 OLED module

Controller + display

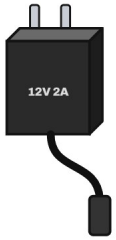
B **1x**



4-channel MOSFETTI

Switches the 12V valves

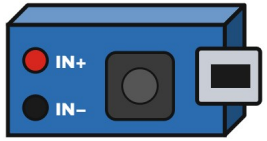
C **1x**



12V 2A power adapter

Main supply

D **1x**



12V → 5V USB converter

Powers the ESP32

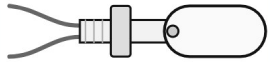
E **1x**



Waterproof DS18B20 + 4.7k

Greenhouse air temperature

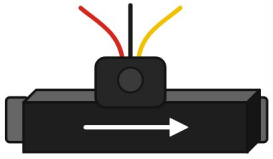
F **2x**



Side-mount float switch

Tank LOW + HIGH

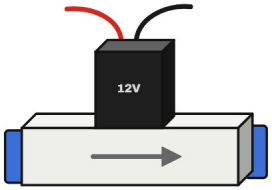
G **1x**



Hall water-flow sensor

Counts litres

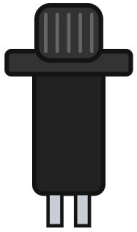
H **3x**



¾" 12V N/C solenoid valve

Water outputs A + B + C

J **4x**



5×20mm fuse holder

F1–F4 protection

K **3x**



Capacitive soil sensor

2 in use + 1 spare

L **1x**



IP55 box 150×110×70

Enclosure

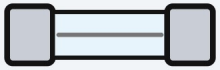
19

pieces in the box

Tick each one off before Step 1

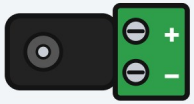
Get these extras first

Not in the PiShop order



1x 2A + 4x 1A

Glass fuses 5×20mm + 1 more fuse holder (F5)



1x

5.5×2.1mm barrel socket to screw terminals



1x

Short USB-A to Micro-USB cable



red + black

Hook-up wire 0.5–0.75 mm²



5x

Female-female Dupont leads



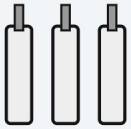
10k + 18k

Resistors for the flow divider



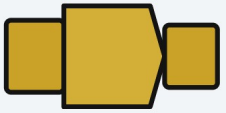
4+

Cable glands, terminal blocks, heat-shrink, ties



1 set

M3 nylon standoffs + screws



to suit

½″ + G¾ fittings, PTFE tape, hose adapters

⚠

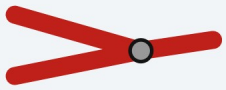
230V rule

No mains wiring in the box. Only the adapter's 12V lead goes in.

Tools



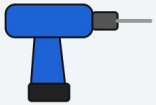
Multimeter



Wire stripper



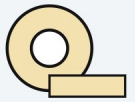
Small screwdrivers



Drill + step bit



Soldering iron

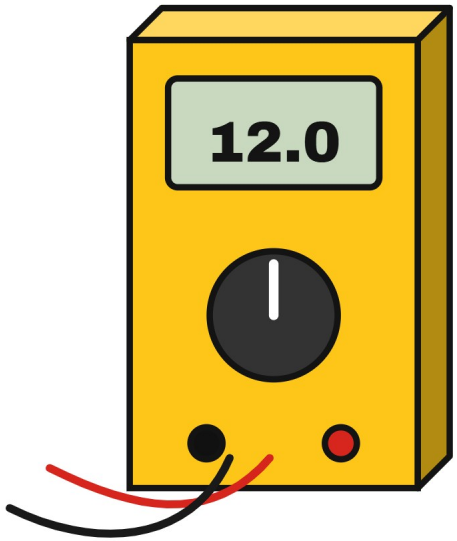


Marker + labels

Wire colours + rules

Use the same colours on every page

RED	+12V power
BLACK	0V / GND
ORANGE	+5V
PURPLE	+3.3V sensor power
GREEN	Analogue signal
YELLOW	Digital / pulse signal
BLUE	Valve control GPIO



⚠ **Never guess polarity**

Trace every red wire from +12V and every black wire to GND before power goes on.

⚠ **Build dry first**

Bench-test all electronics on a table. Plumbing and tanks come after the tests pass.

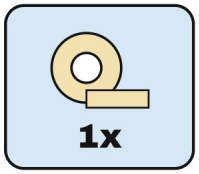
⚠ **Adult check points**

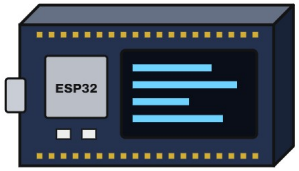
Drilling, soldering the MOSFETTI, stripping cable and the first live 12V power-up.

⚠ **230V stays outside**

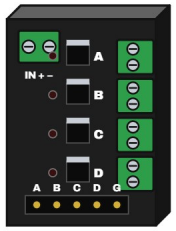
The adapter plugs into a proper wall outlet outside the box. Only 12V goes in.

1 Unpack, label and sort

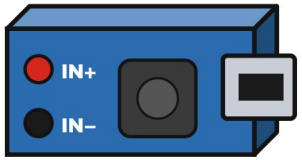




A · ESP32



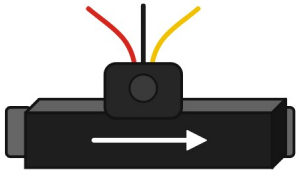
B · MOSFETTI



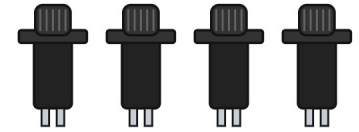
D · 5V

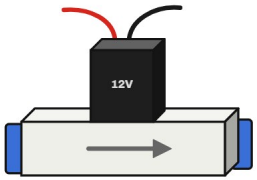


E · AIR TEMP

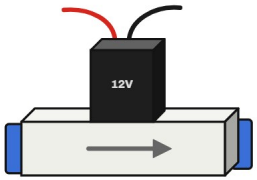


G · FLOW

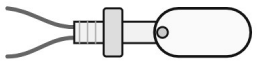




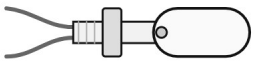
H · VALVE A



H · VALVE B



F · FLOAT LOW



F · FLOAT HIGH



1 · ALLEY POTS



2 · BED

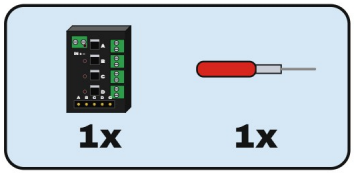


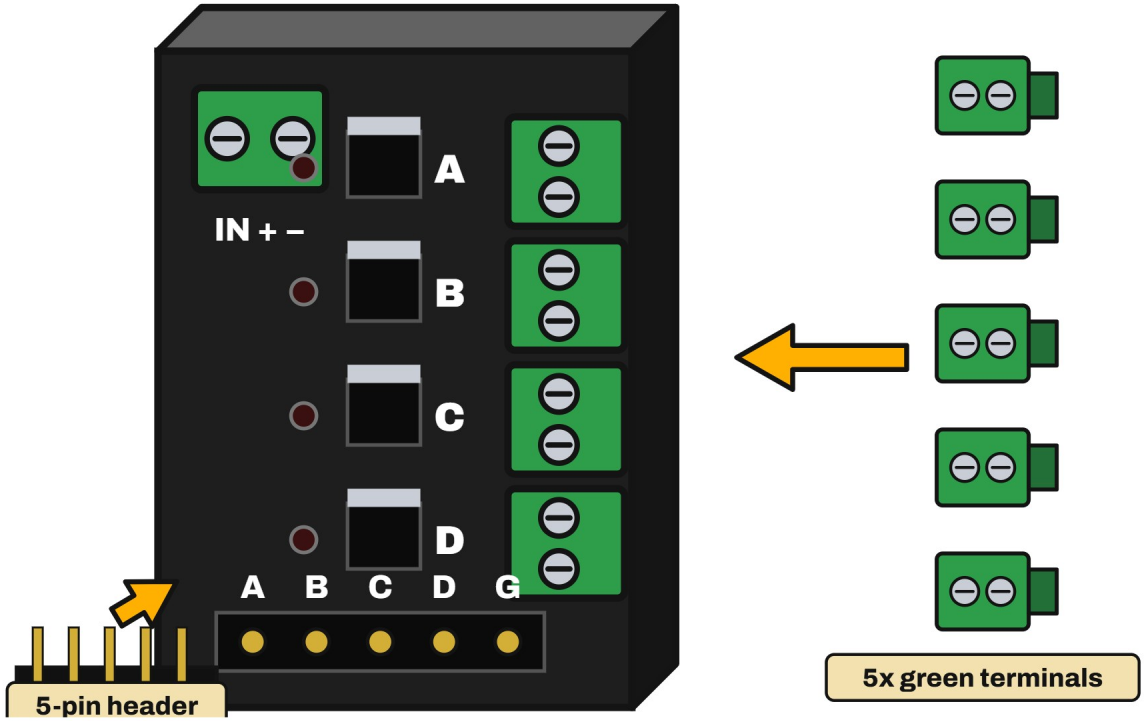
SPARE

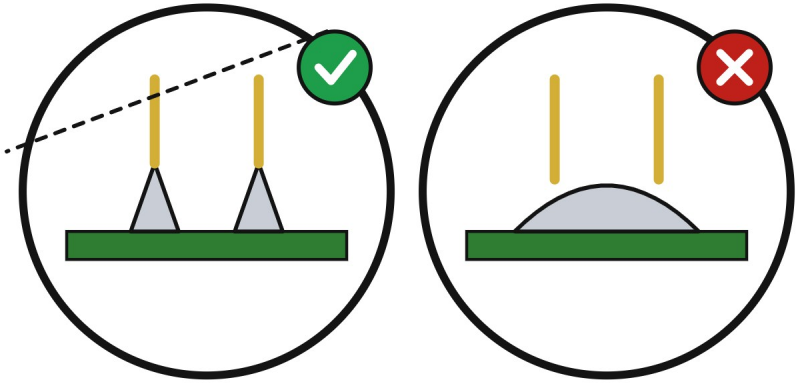
1. Lay every part on a clean, dry table. Keep the 12V adapter unplugged.
2. Stick a masking-tape label on every part exactly as shown. Third valve: H · VALVE C. Fifth fuse holder: F5 1A.
3. Put the tiny 4.7k resistor from the DS18B20 bag in its own bag: DS18B20 PULL-UP.
4. Do not wire anything yet.

CHECK: every duplicate part has its own label.

2 Solder the MOSFETTI kit





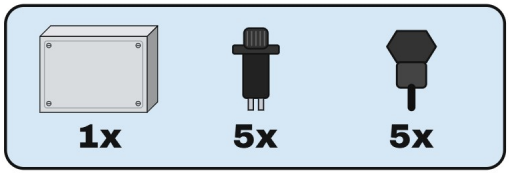


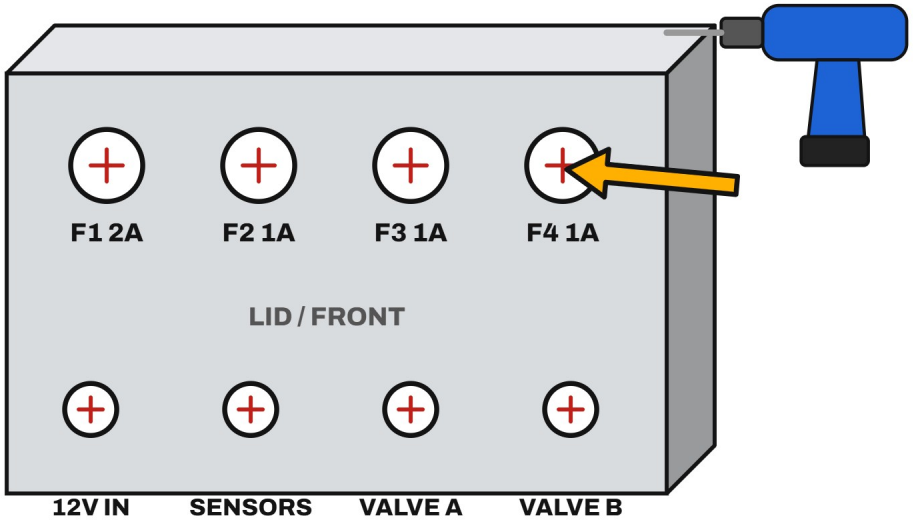
1. Solder the 5-pin header into A, B, C, D, GND. Keep it straight.
2. Fit the 5 green screw terminals. Wire openings face **outward**.
3. Solder every pin, then let the board cool.
4. Inspect: shiny cone joints, no solder bridges between pads.

⚠ **MOSFETTI rule:** Low-voltage DC only. Its A–D inputs accept 3V or 5V logic, so the ESP32 drives it directly.

✓ **CHECK: terminals face outward and the header is straight.**

3 Drill the box

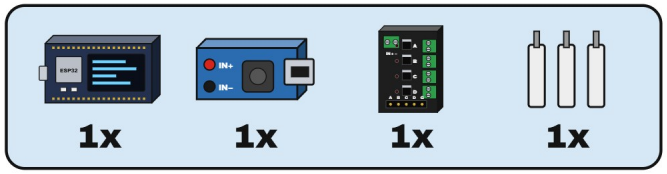


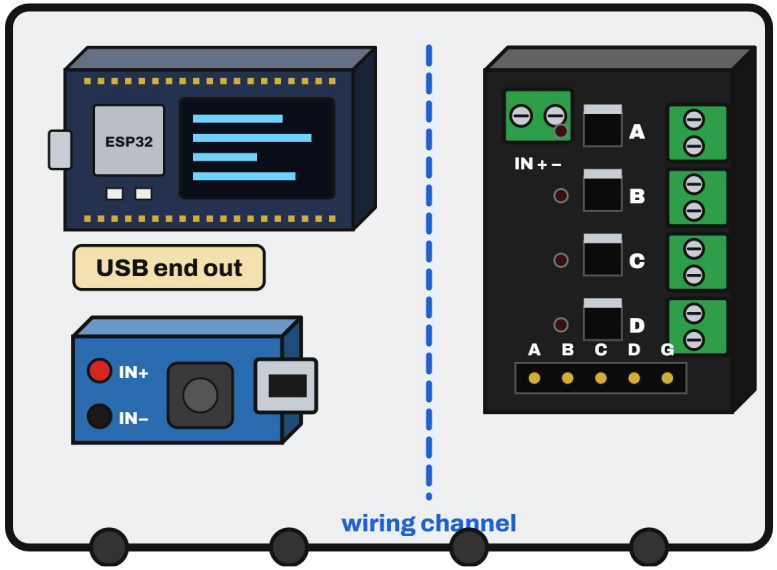


1. Mark 5 fuse holes on the lid where labels can be read (F5 = Valve C).
2. Mark cable-gland holes along the bottom edge, plus one more for VALVE C.
3. Drill slowly. Remove every plastic chip, then dry-fit glands and holders.

CHECK: no plastic chips, no sharp edges.

4 Mount the boards

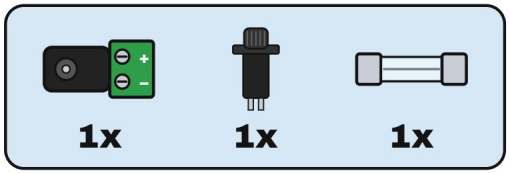


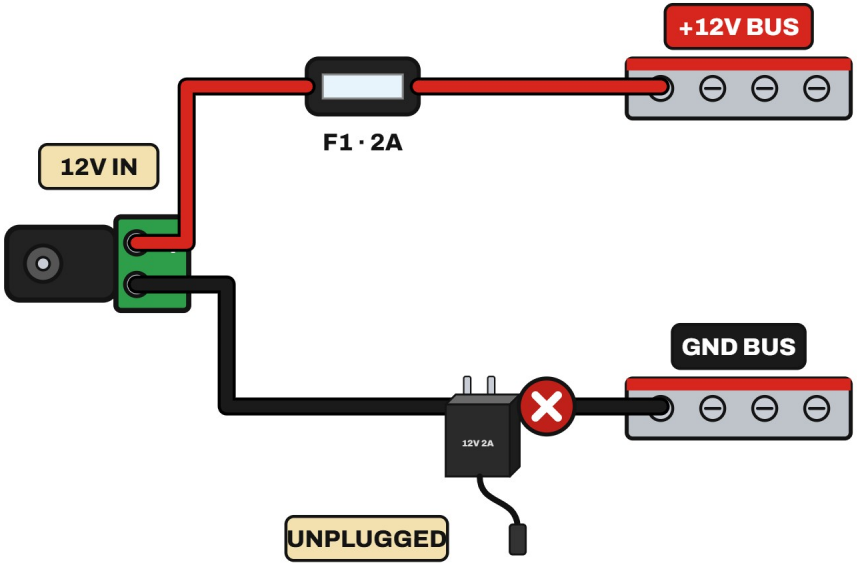


1. ESP32 upper left, Micro-USB reachable.
2. 5V converter below it, with air around.
3. MOSFETTI on the right, terminals facing you. Nylon standoffs only.

CHECK: the lid closes without touching boards or cable.

5 12V input + main fuse

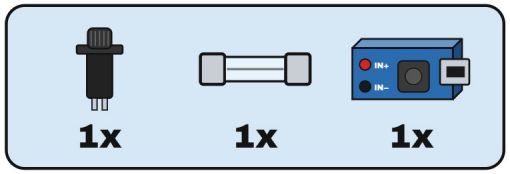


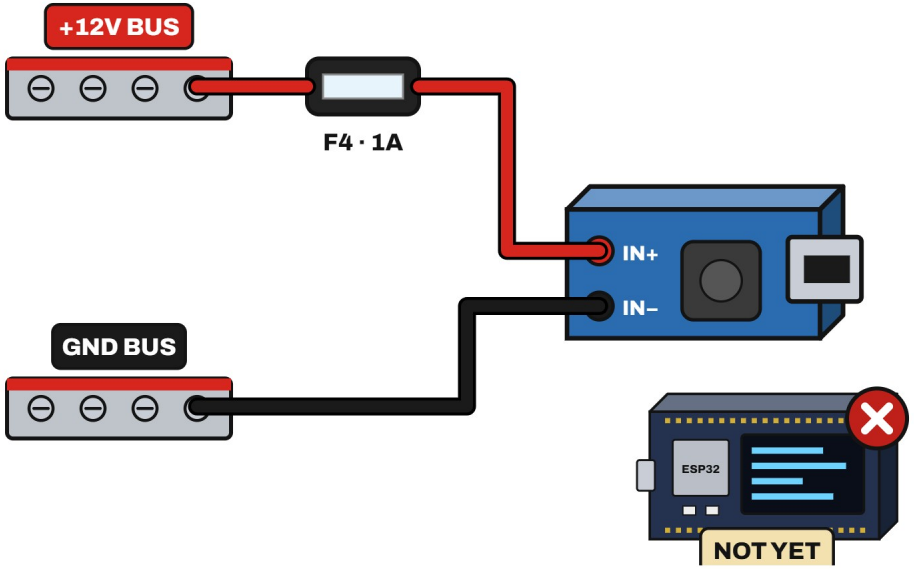


1. Fit a 2A fuse in F1. Barrel centre = +12V, sleeve = GND.
2. +12V goes through F1 to the +12V BUS.
3. GND goes straight to the GND BUS. GND is never fused.

CHECK: continuity test: centre only reaches red, sleeve only black.

6 5V branch for the ESP32

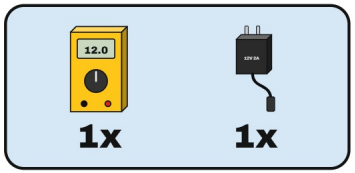


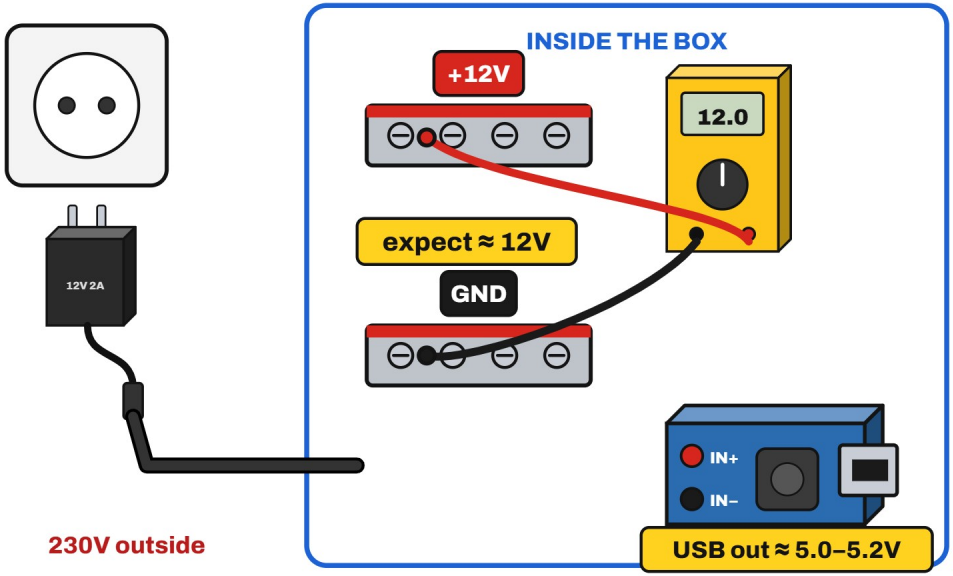


1. Fit a 1A fuse in F4.
2. +12V BUS → F4 → converter IN+. GND BUS → IN–.
3. Leave the ESP32 unplugged: measure first.

CHECK: IN+ and IN– are not swapped. The converter is not reverse-proof.

7 First power-up: measure

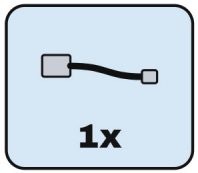


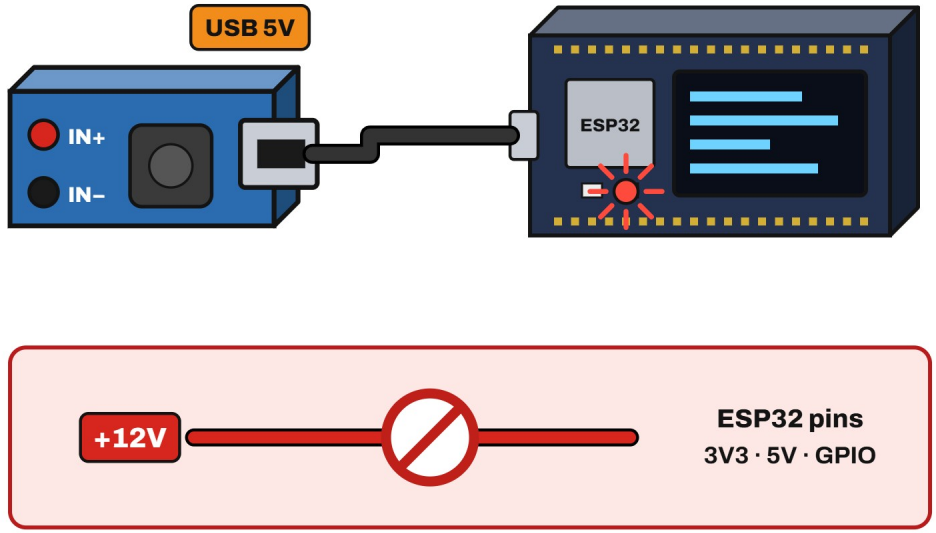


1. Meter on DC volts. Plug the adapter into the barrel socket, then the wall.
2. +12V BUS to GND: about 12V. Converter USB output: about 5.0–5.2V.
3. Reading wrong, zero or reversed? Unplug at once and recheck.

CHECK: both readings pass before you go on.

8 Power the ESP32 by USB

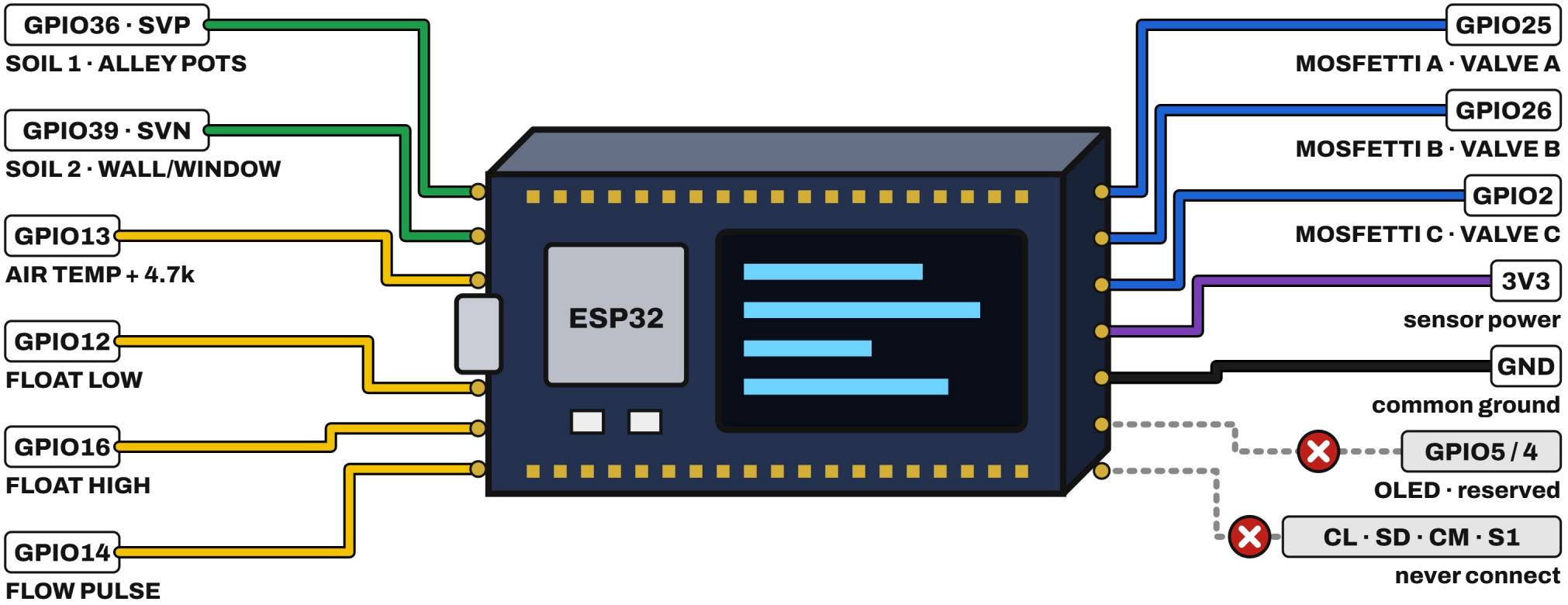




1. Unplug the adapter. Run the short USB lead from converter to ESP32.
2. Plug the adapter back in. The power LED lights. A blank OLED is normal.
3. After 30 s touch both boards: warm is fine, hot is not.

CHECK: the ESP32 powers up and nothing gets hot.

9 The ESP32 pin plan · your 26-pin board

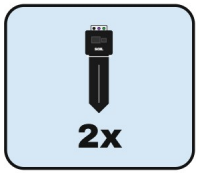


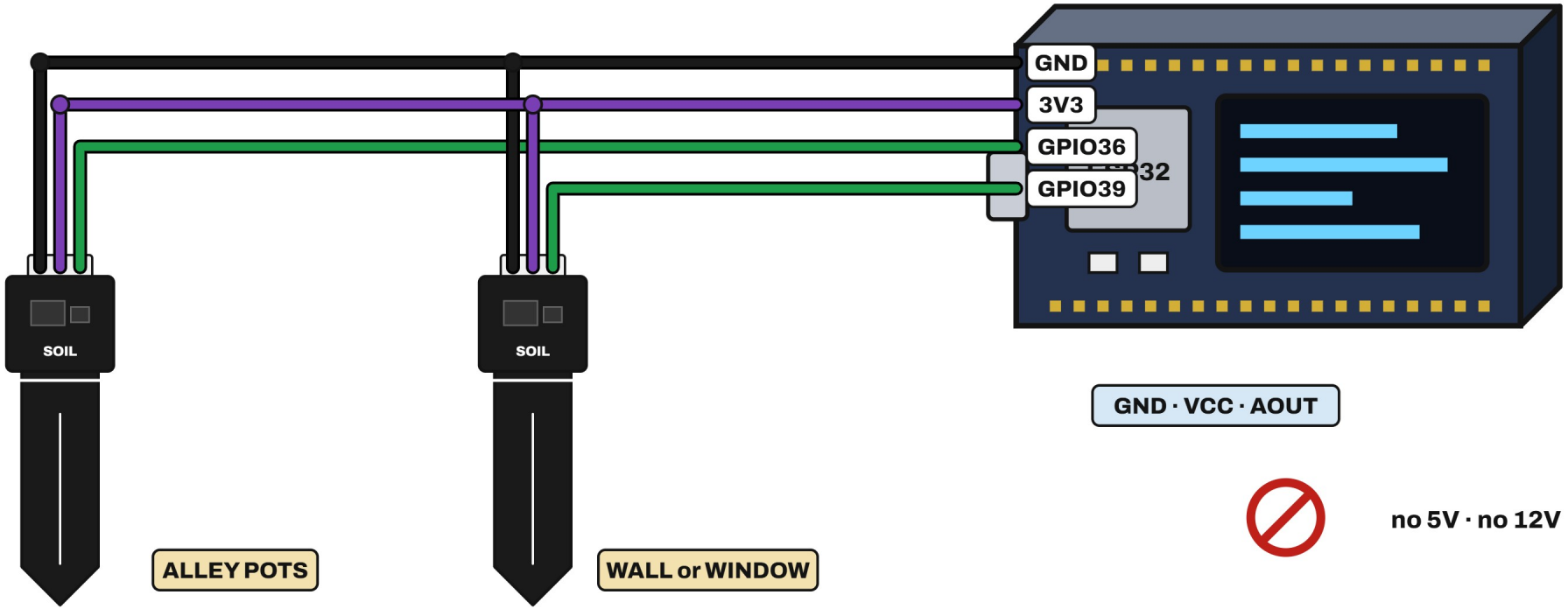
1. Print this page and keep it next to the box while wiring.
2. Write the GPIO number on a tape flag on every wire.

Why these pins? This 26-pin board has only two analogue pins that work with Wi-Fi on (SVP, SVN), so it reads two soil sensors. GPIO12 must be low at start-up: only a float switch (to GND) goes there. GPIO25, 26 and 2 drive MOSFETTI A, B and C (this board has no 27, 32 or 33; GPIO2 stays low at start-up, so Valve C stays shut while it boots). GPIO5/4 run the screen. CL, SD, CM, S1 run the memory chip. Also leave 0, 15, TX and RX free.

CHECK: every wire you add matches this map.

10 Connect the two soil sensors

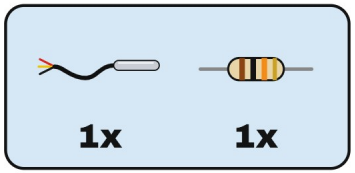


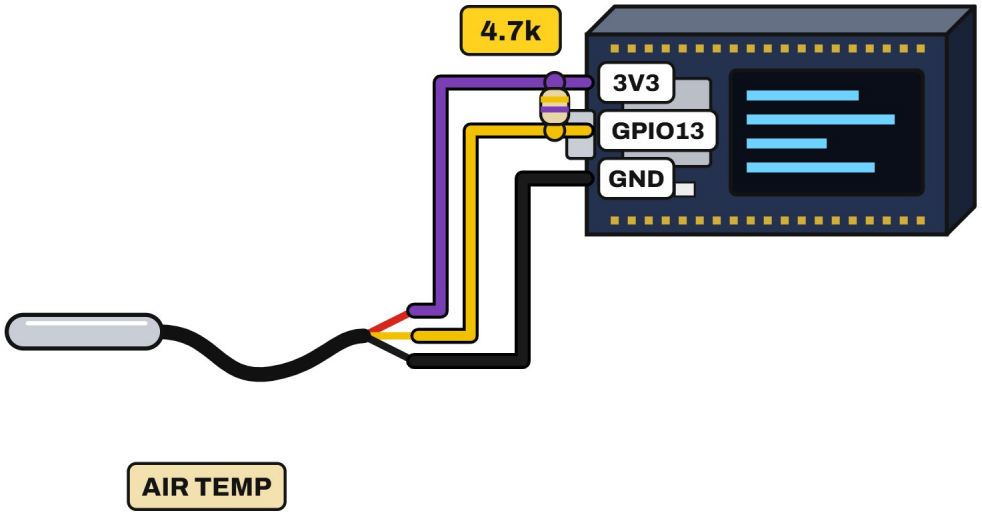


1. Unplug power. Your boards read GND · VCC · AOUT: black, red, yellow on the factory lead.
2. VCC → ESP32 3V3 (purple). GND → GND (black). Both sensors share these rails.
3. Alley Pots AOUT → GPIO36 (SVP). The other bed's AOUT → GPIO39 (SVN).
4. The third sensor stays spare. Only the coated blade goes in the soil.

CHECK: no soil sensor touches +5V or +12V.

11 Air-temperature probe

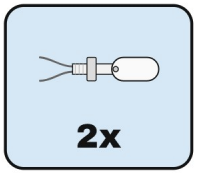


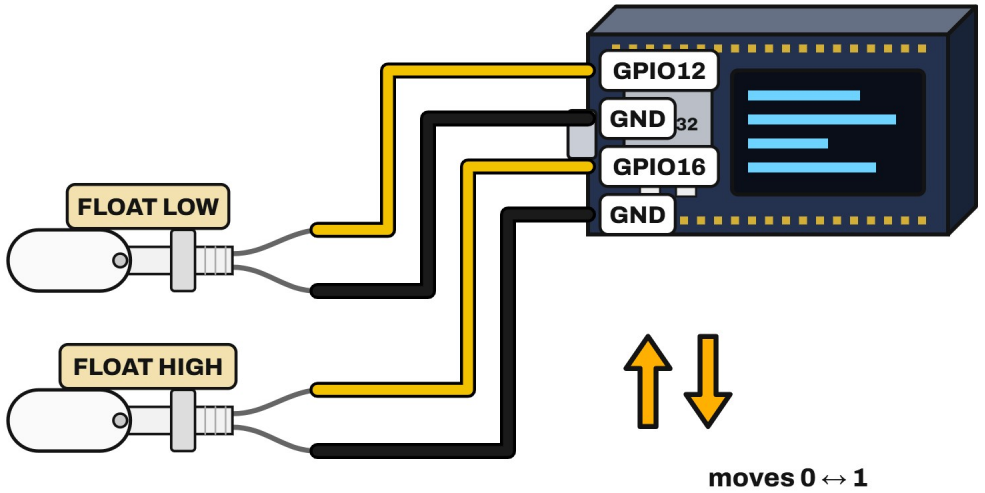


1. Red → 3V3. Black → GND. Yellow/white DATA → GPIO13. Leave a bare shield wire unconnected.
2. The 4.7k resistor bridges DATA to 3V3. Never on GPIO12: it would stop the board starting.
3. Label the steel probe AIR TEMP. It will hang in the greenhouse air (Step 18).

CHECK: the 4.7k goes DATA to 3V3, never DATA to GND.

12 Tank float switches

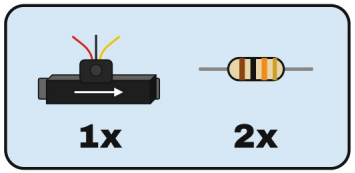


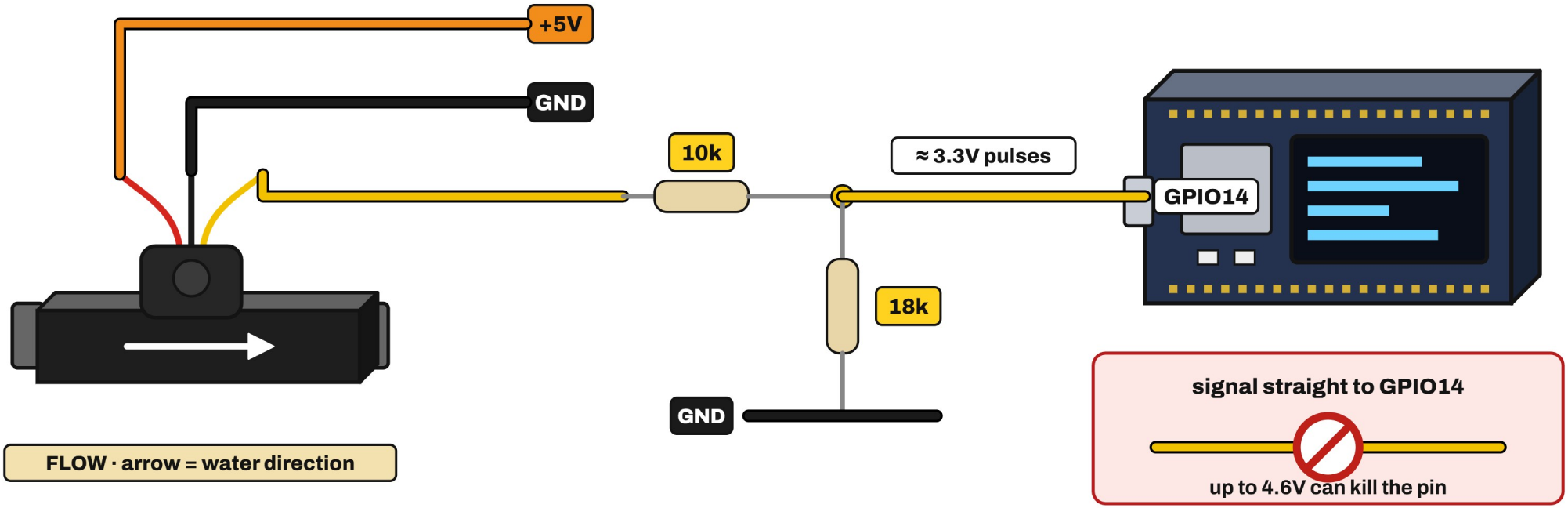


1. A float is a plain switch: no + or –, no power needed.
2. LOW: one wire → GPIO12, other → GND. HIGH: one wire → GPIO16, other → GND.
3. GPIO12 must be low at start-up. A float to GND is safe there; nothing else goes on 12.

CHECK: each float sits only between its GPIO pin and GND.

13 Water-flow sensor with a safe divider

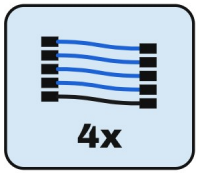


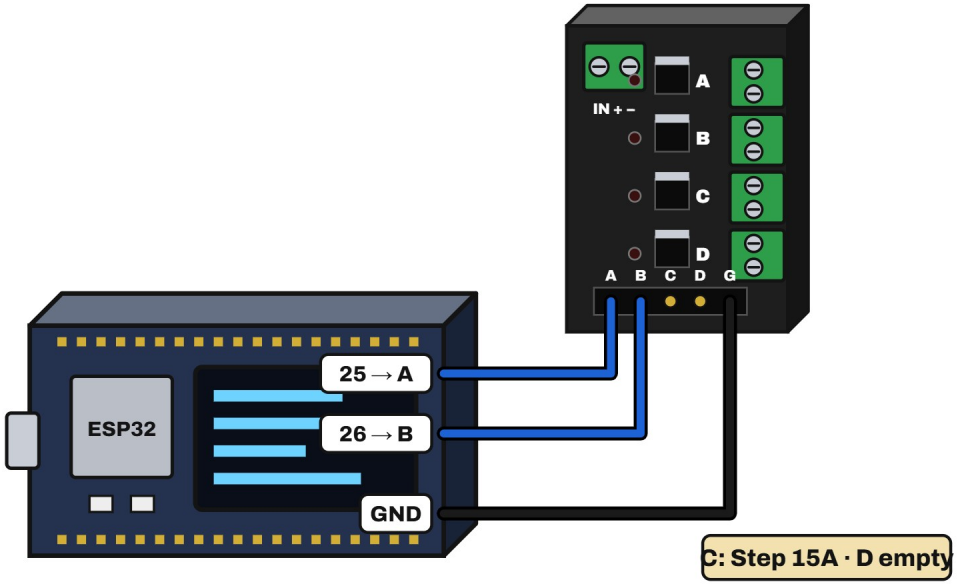


1. Unplug power. Sensor red → +5V (orange), black → GND.
2. Build the divider: signal → 10k → junction → 18k → GND.
3. Junction → GPIO14. Never wire the signal straight to the pin.
4. Cover both resistors with heat-shrink so nothing can short.

CHECK: GPIO14 connects to the divider junction only.

14 ESP32 → MOSFETTI control

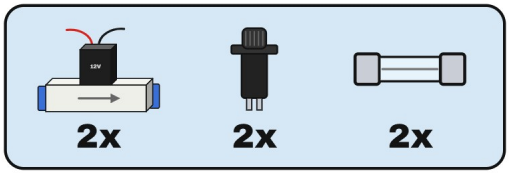


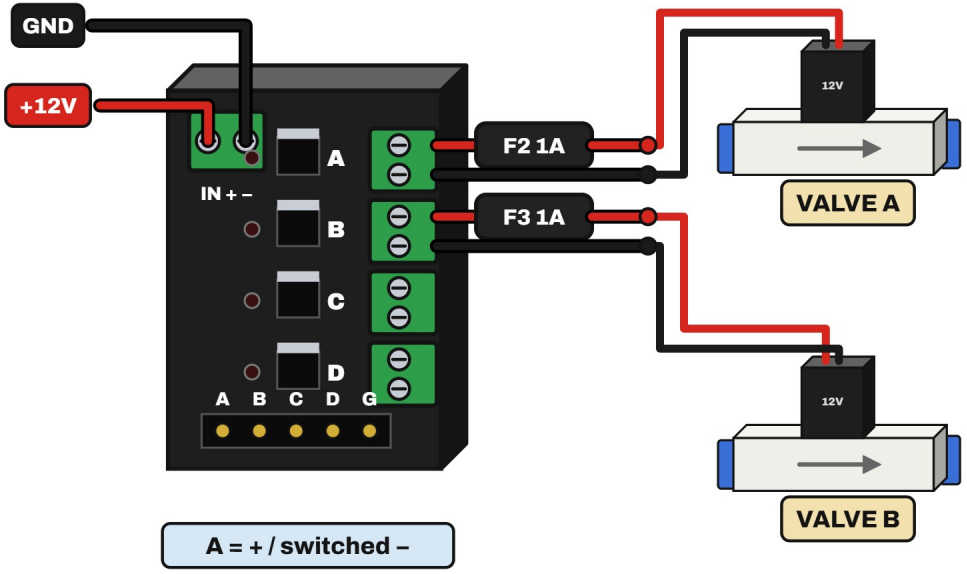


1. ESP32 GND → MOSFETTI header GND. This link is required.
2. GPIO25 → A, GPIO26 → B. GPIO2 → C is added in Step 15A. Leave D empty.
3. The test code holds A, B and C LOW at boot, so every valve stays shut.

CHECK: ESP32 GND and MOSFETTI GND are joined.

15 12V to MOSFETTI and valves

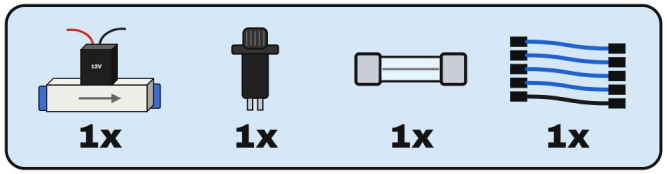


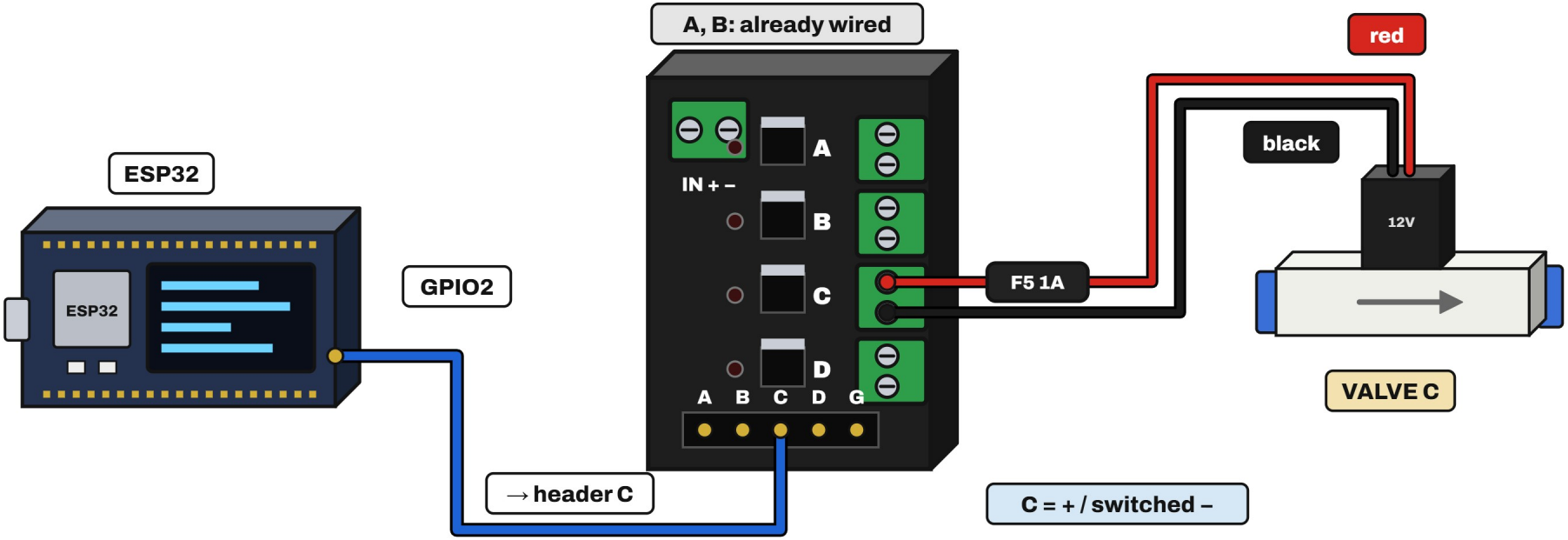


1. +12V BUS → MOSFETTI IN +. GND BUS → IN GND.
2. Out A: + → F2 (1A) → Valve A red. Switched terminal → Valve A black.
3. Repeat Out B through F3 to Valve B. Valve C follows in Step 15A. Each valve draws about 0.42A.

CHECK: each valve has its own 1A fuse. MOSFETTI max is 16V DC.

15A Add Valve C · third MOSFETTI channel

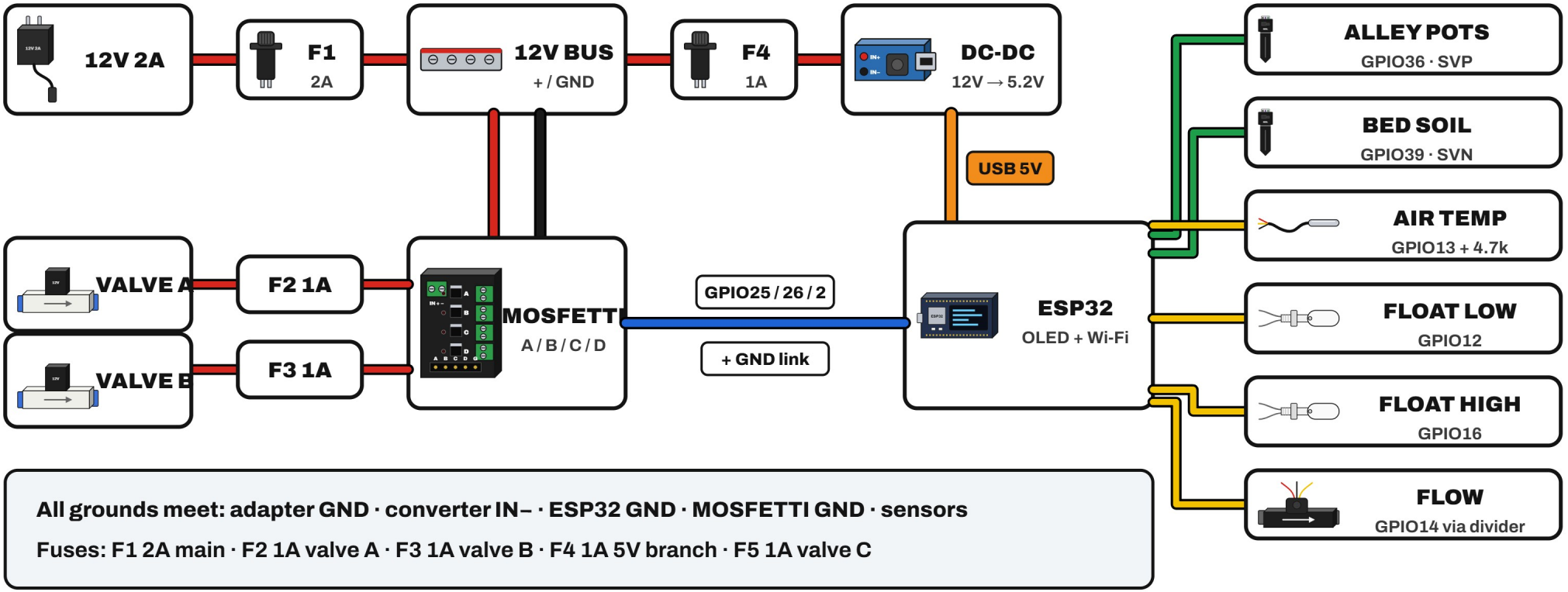




1. Unplug the 12V adapter. Fit the fifth fuse holder, F5, on the lid with a 1A fuse.
2. Find the pin printed 2 on the ESP32 (right header, between 0 and 14). Dupont lead: GPIO2 → MOSFETTI header C. The GND link from Step 14 already covers C.
3. Out C: + → F5 (1A) → Valve C red. Switched terminal → Valve C black. Its cable leaves through its own gland.
4. Three valves at about 0.42A each is about 1.3A. The 12V 2A adapter and the 2A F1 still cover it. The ESP32's blue LED may light while Valve C is open: that's normal.

CHECK: GPIO2 goes only to C, and Valve C has its own 1A fuse (F5).

16 The complete model: check every wire

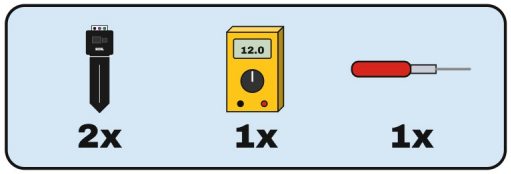


1. Unplug power. Trace every wire in the box against this page. Valve C (Step 15A) is wired like A and B, through F5.
2. Highlight each line on the printout only after you have touched it in the box.

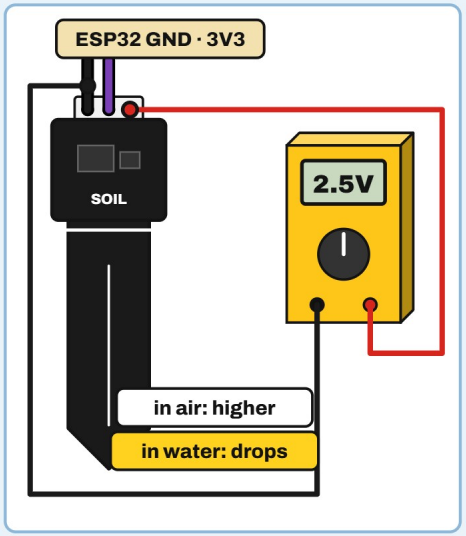
CHECK: no bare strands, no loose screws, no 12V near sensor or GPIO wiring.

16A

Prepare the soil sensors · do this before Step 18

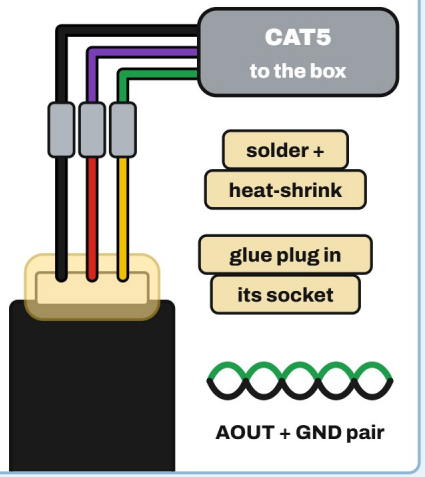


1 Check it works



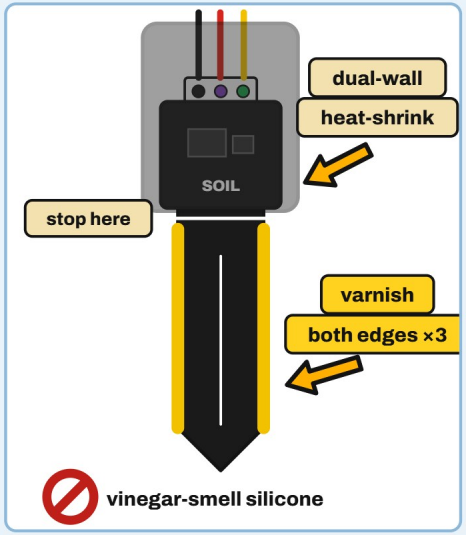
Power the sensor from the ESP32 (USB only). Meter red on AOUT, black on GND. Dip the blade in water: the voltage must drop.

2 Extend the lead



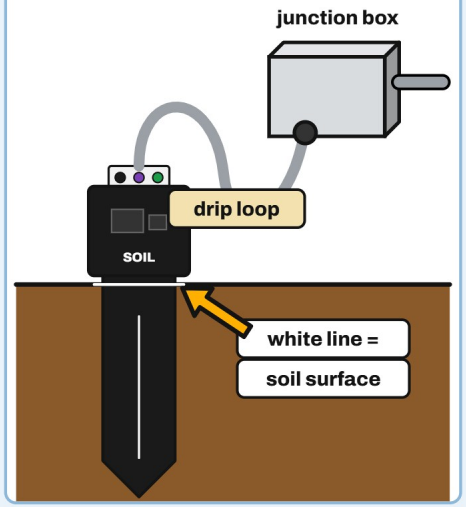
Keep the factory lead plugged in. Cut off the black end plug and join each wire to CAT5, AOUT and GND on one twisted pair.

3 Seal it



Shrink adhesive-lined tube over the plug and electronics, down to the white line. Varnish both cut edges of the blade.

4 Install

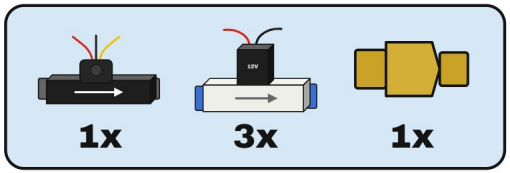


Push in no deeper than the white line. Joints and splits live in a small sealed box above the soil.

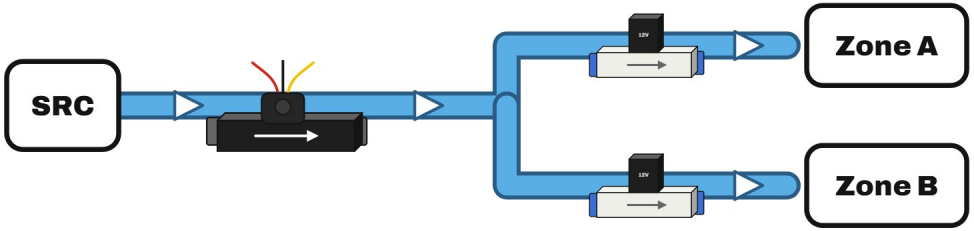
Extra to buy: outdoor CAT5/CAT6 cable · dual-wall (adhesive) heat-shrink about 12 mm · clear nail varnish or conformal coat · a small IP65 junction box for the bed end · neutral-cure silicone or epoxy if you prefer potting.

CHECK: every sensor still reads after sealing, and nothing is bare above the white line. Calibrate (Step 23) only after sealing.

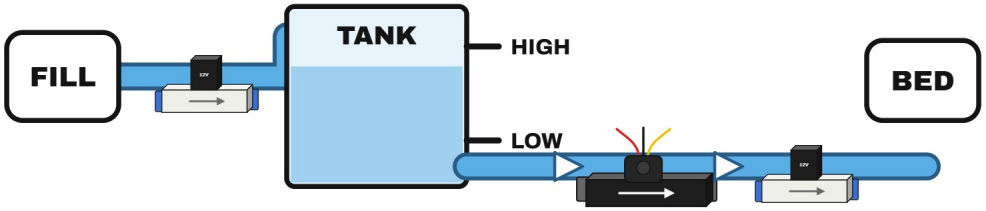
17 Build the water side



OPTION A · two zones



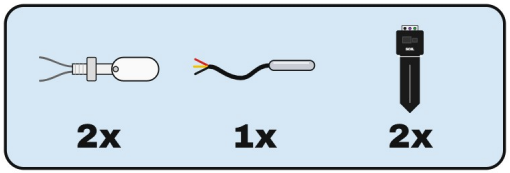
OPTION B · tank fill + irrigation

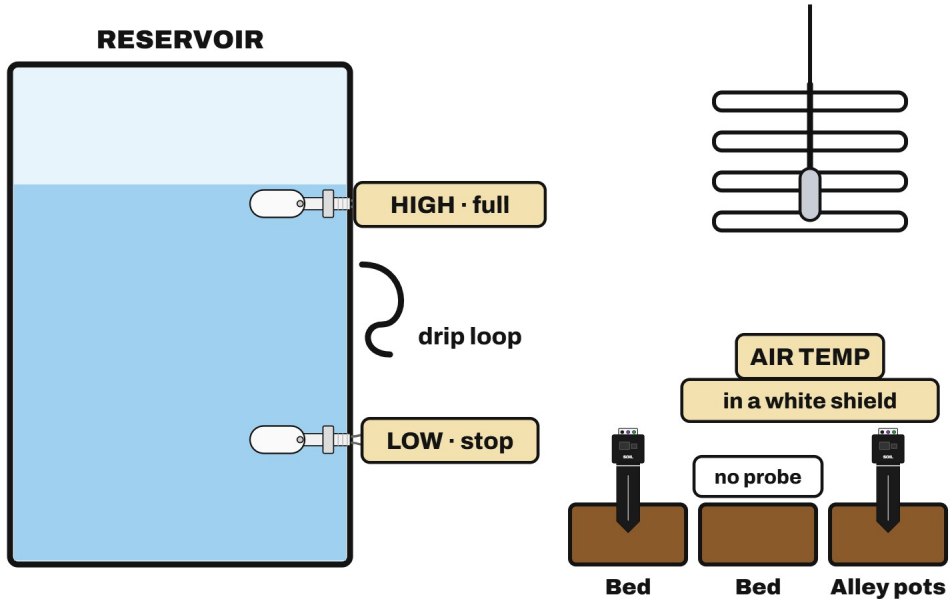


1. Only after every electrical bench test has passed.
2. Pick a layout. Valve C adds a third zone (A) or a second bed (B). Flow arrows on sensor and valves point the way water moves.
3. Valves need 0.02–0.8 MPa. A low tank on gravity may need a pump.

CHECK: arrows match the water direction. Leak-test before the box goes near.

18 Place floats and probes

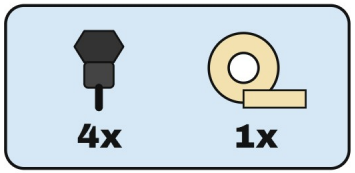


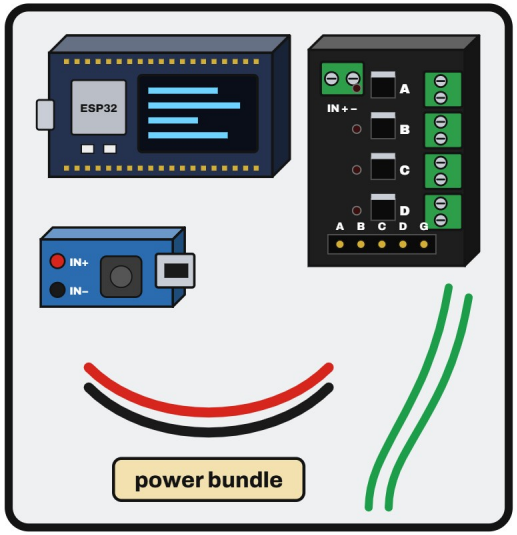


1. LOW float at the minimum level where watering must stop. HIGH at normal full.
2. Air-temp probe: tip-down in a white sun shield at plant height, away from doors, vents and spray. Soil probes: Alley Pots + the faster-drying bed, in the root zone.
3. Every cable gets a drip loop before it enters the box.

CHECK: plumbing is pressurised and leak-free before power goes on.

19 Finish the box





LID LABEL

F1 2A · F2 A 1A

F3 B · F4 5V · F5 C 1A

36 POTS · 39 BED

13 TEMP · 14 FLOW

12 LOW · 16 HIGH

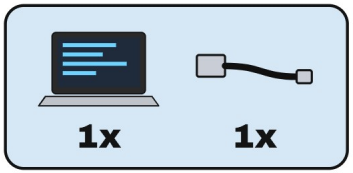
25 A · 26 B · 2 C

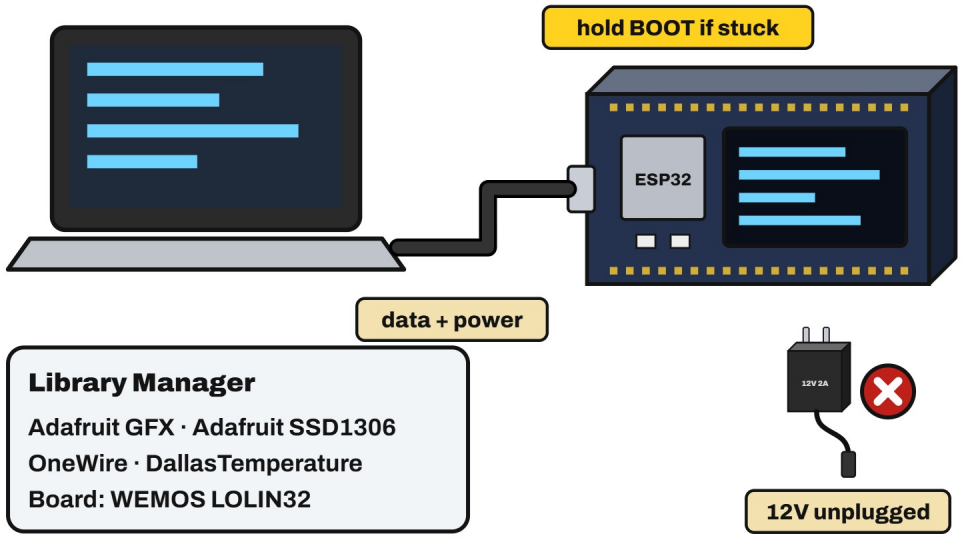
5/4 OLED · DRY / WET

1. Bundle 12V power wires away from the soil-sensor wires. Tug-test every terminal.
2. Stick the pin map, fuse ratings and valve names inside the lid.
3. Glands grip the cable jacket. Leave the lid open until all tests pass.

CHECK: the box can be opened later without cutting a wire.

20 Load the test sketch





Library Manager

Adafruit GFX · Adafruit SSD1306

OneWire · DallasTemperature

Board: WEMOS LOLIN32

1. Install Arduino IDE, the ESP32 boards package and the four libraries.
2. Unplug the 12V adapter and the converter USB. Plug the ESP32 into the PC.
3. Open greenhouse_hardware_test_v3.ino, change PIN_VALVE_C from 27 to 2, and upload. Full walkthrough: Step 20A.

CHECK: Arduino IDE says "Done uploading". Never PC USB + converter USB together.

20A

Program the ESP32 · the full walkthrough for Step 20

Do this once on the PC. Every later upload is just steps 4 to 6.

1 Install Arduino IDE

arduino.cc **Software** **IDE 2.x**

Download and install it on the PC. Open it once so it finishes setting up.

2 Add the ESP32 boards

```
https://espressif.github.io/arduino-esp32/package_esp32_index.json
```

File › Preferences: paste this into Additional boards manager URLs. Then Boards Manager: install **esp32 by Espressif Systems**.

3 Add the four libraries

Adafruit GFX **Adafruit SSD1306**

OneWire **DallasTemperature**

Tools › Manage Libraries. Search each name and install. Say yes to any extra libraries it offers.

4 Pick board and port

Board: WEMOS LOLIN32

Port: COM… / ttyUSB…

No port? Use a data USB cable, then install the CP210x or CH340 driver printed near the USB socket.

5 Open and upload

greenhouse_hardware_test_v3.ino

Upload →

Wait for **Done uploading**. Stuck on Connecting…? Hold BOOT, tap EN. Still failing: Tools › Upload Speed › 115200.

6 Talk to it

```
Soil: 2870, 3012 | Temp 21.8 | LOW 1 HIGH 1 | Flow 0
> C
Testing Valve C
Valve OFF
```

Serial Monitor at 115200. Type A, B or C + Enter to open that valve 2 s. R resets the flow count.

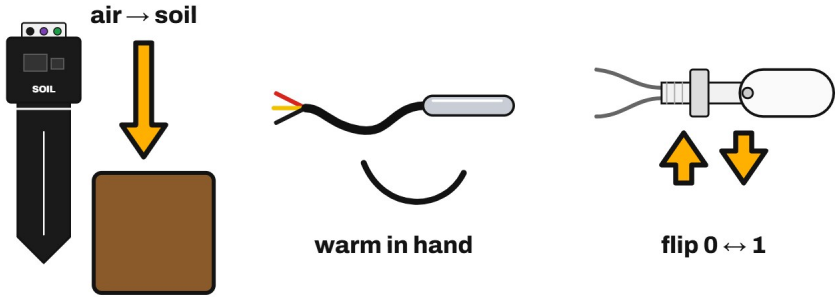
USB rule: 12V adapter unplugged and converter USB out before the PC cable goes in. Never both USB leads at once. Valve tests (Step 22) add 12V while the PC stays connected.

CHECK: the IDE says Done uploading, and the Serial Monitor prints one line every second.

21 Test every sensor

Serial Monitor · 115200

```
Soil: 2870, 3012 | Temp 21.8 | LOW 1 HIGH 1 | Flow 0
Soil: 1840, 3008 | Temp 21.9 | LOW 0 HIGH 1 | Flow 0
Soil: 1836, 2999 | Temp 22.0 | LOW 0 HIGH 1 | Flow 48
one line every second
```



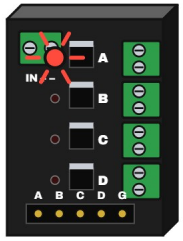
1. 12V stays unplugged. The PC powers the ESP32 and sensors.
2. Soil: air, then damp soil. The number changes. Temp: warm it in your hand. Floats: lift by hand.
3. Flow: pulses rise only when water runs through.

CHECK: both soils, temperature, both floats and flow respond.

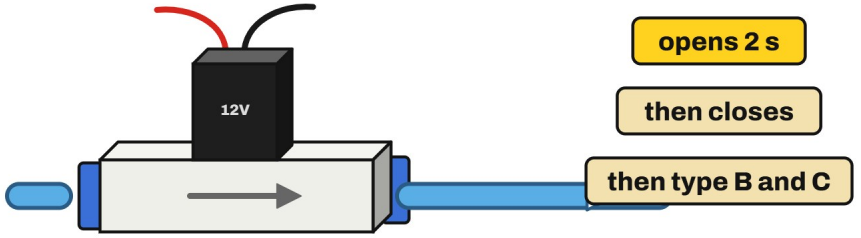
22 Test Valves A, B and C

Serial Monitor · 115200

```
> A
Testing Valve A
Valve OFF
```



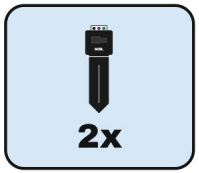
A lights

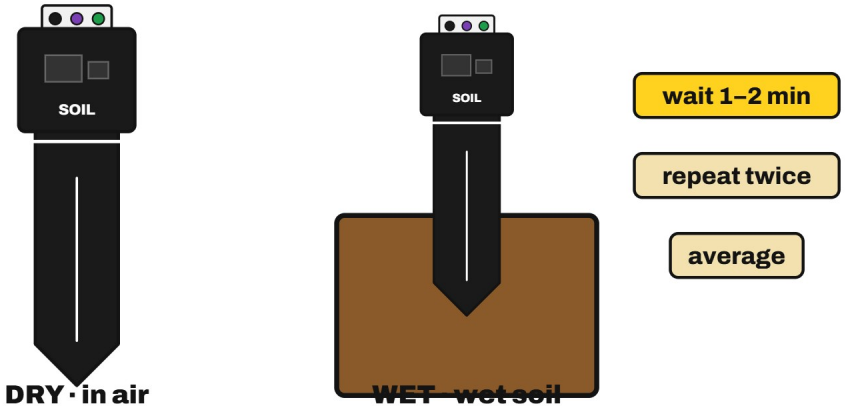


1. Valves plumbed, leak-free, an adult watching. ESP32 on PC USB, converter USB off, then plug in 12V.
2. Type A + Enter: LED A lights, Valve A opens 2 s, closes. Type B for Valve B, C for Valve C.
3. LED on but no click? Unplug. Check fuse, 12V at the output, wiring, pressure.

CHECK: each letter moves only its own valve, and every valve closes.

23 Calibrate the soil sensors

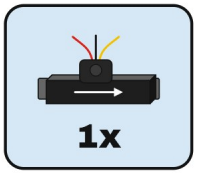


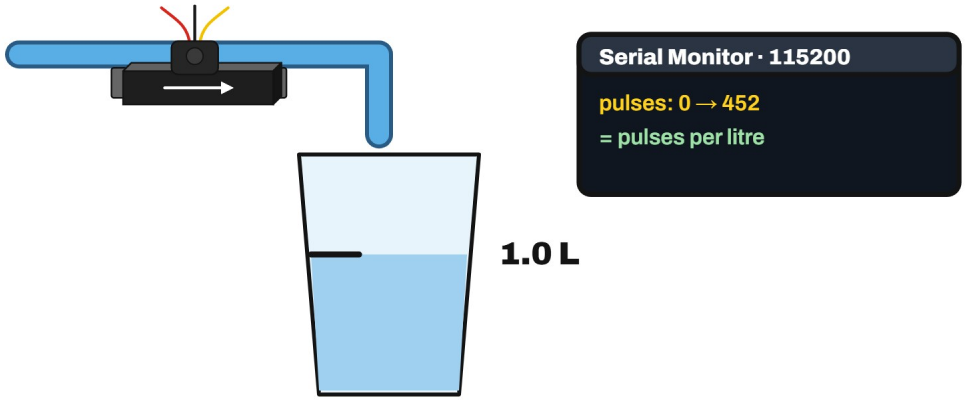


1. Write each reading in air (DRY) and in fully wet bed soil (WET). Drier reads higher.
2. Never submerge the top electronics. Recalibrate after soil changes.

Sensor	DRY raw	WET raw
1 · Alley Pots		
2 · Bed		

24 Calibrate the flow sensor





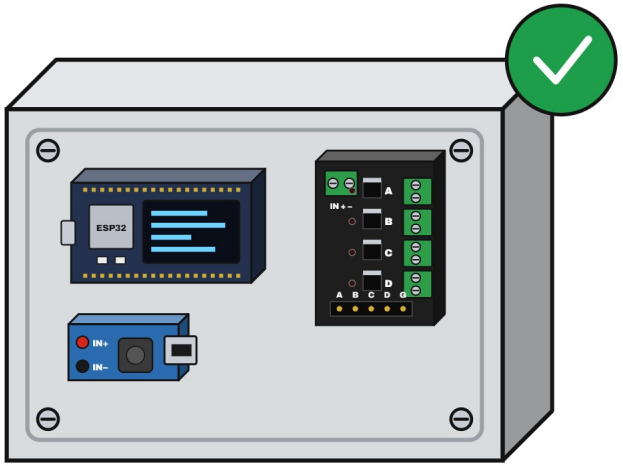
1. Note the pulse count, run exactly 1.0 L into a jug, note it again.
2. Do it three times. The average is your PULSES_PER_LITRE.

Run	Pulses for 1.0 L
1	
2	
3	
Average	

25 First complete system test

ESP32 on PC USB, converter USB off, 12V plugged in. Fire each valve for 2 seconds and watch the box and the plumbing.

- [ ] **POWER: 12V and 5V correct**
- [ ] **FUSES: F1 2A · F2 1A · F3 1A · F4 1A · F5 1A**
- [ ] **SENSORS: all respond**
- [ ] **FLOATS: LOW and HIGH change**
- [ ] **FLOW: pulses when water moves**
- [ ] **VALVE A: only A opens, right outlet**
- [ ] **VALVE B: only B opens, right outlet**
- [ ] **VALVE C: only C opens, right outlet**
- [ ] **LEAKS: none**
- [ ] **HEAT: normal after the run**



Builder _____

Date _____

12V measured _____

5V measured _____

⚠ **Rule:** If any line fails, fix it first. Automation never hides unreliable hardware. When every box is ticked, the hardware is ready for the Raspberry Pi software.